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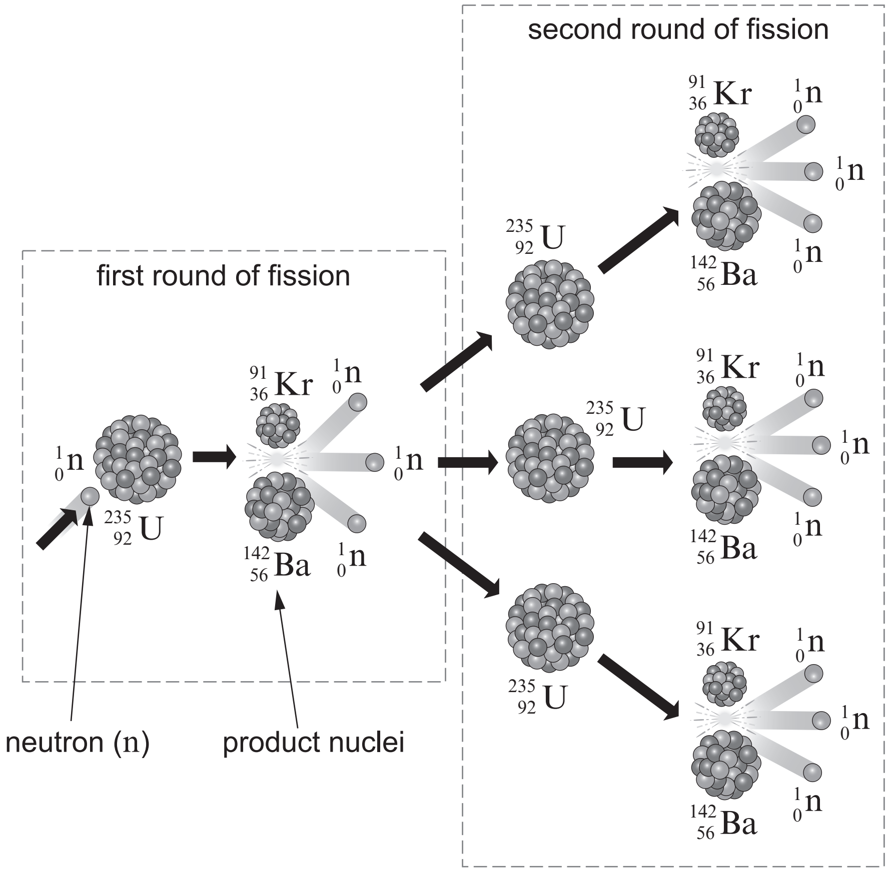
5. The diagram shows one possible chain reaction in the fission of uranium–235 (${}^{235}_{92}\mathrm{U}$) by a single neutron (${}^{1}_{0}\mathrm{n}$).
The product nuclei, krypton (Kr) and barium (Ba), are radioactive.



During the first round of fission, 2 product nuclei are created and 3 neutrons are released.

(a) Tick (✓) the **three** boxes next to the correct statements below. [3]

One neutron splits to create three other neutrons.	☐
The proton numbers of barium and krypton add to give the proton number of uranium-235.	☐
Nuclear fission takes in energy from the nucleus.	☐
During the second round of fission, another 9 neutrons are released.	☐
During the second round of fission, another 6 product nuclei are created.	☐
An atom of uranium-235 has 92 neutrons in its nucleus.	☐

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(b) Use words from the box to complete the following sentences about a nuclear reactor. Each word may be used once, more than once or not at all.

absorb	**one**	**two**	**three**	**fuel**	**moderator**
emit	**concrete**	**reactor**	**control**		

(i) Neutrons are slowed down by the .... so uranium nuclei can .... them. [2]

(ii) .... rods absorb fission neutrons so only .... of these neutrons goes on to produce further fission. [2]

(iii) In an emergency, ... rods are dropped fully into the nuclear .... [2]

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